

ChronoGraph: A Real-World Graph-Based Multivariate Time Series Dataset

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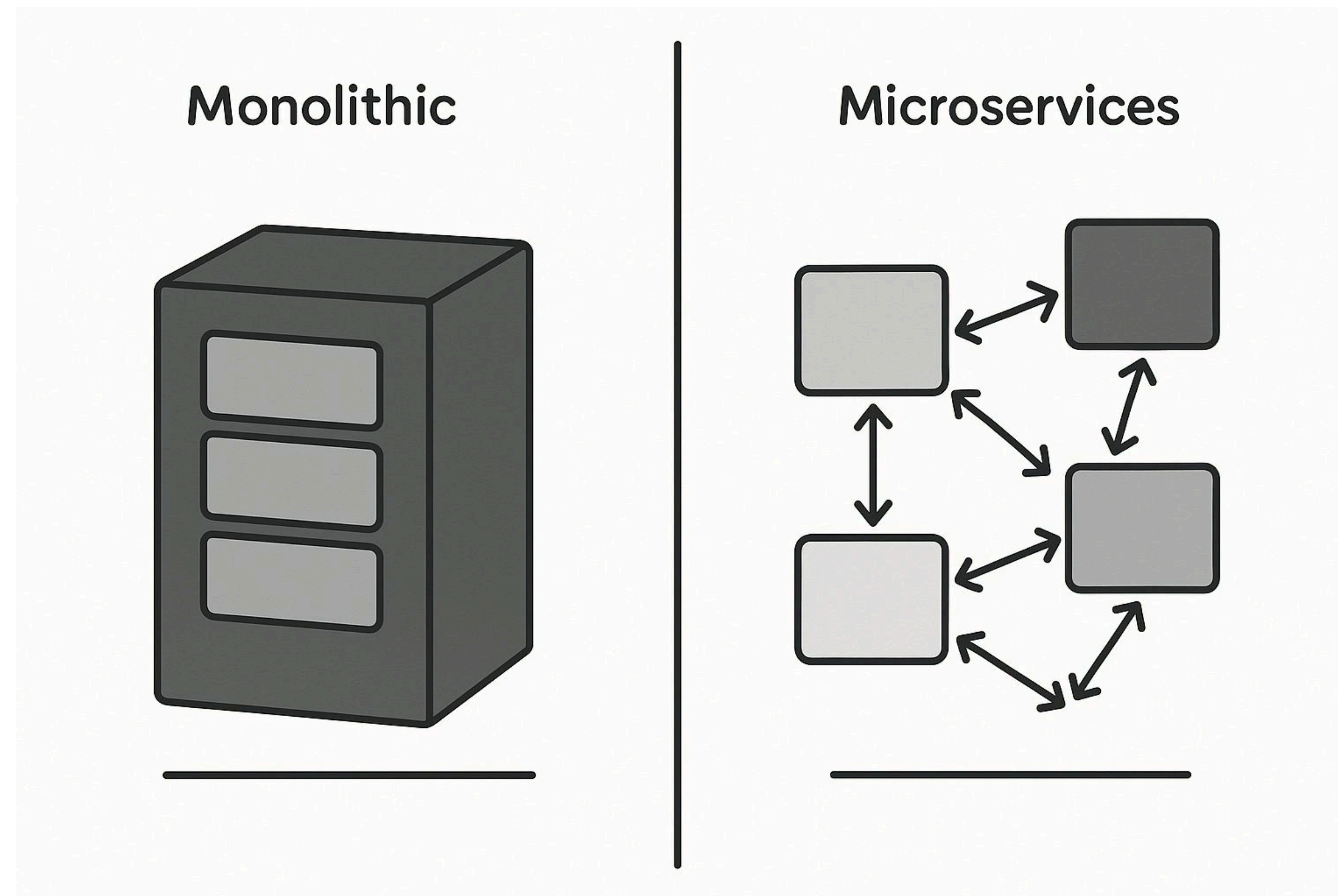
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MOTIVATION

- Microservices are at the base of the world's biggest platforms.
- Forecasting service telemetry is crucial for the reliability of such systems.



What do we want?

- Forecasting monitoring system that:
 - Takes into account the state of the entire system.
 - Can forecast and predict anomalies/system disruptions.
 - Allows us to prevent, not react.
 - Most current methods used in production are heuristic-based.

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Existing benchmarks did meet our needs.

MOTIVATION

Existing benchmarks (traffic, air quality, SWaT [1], WADI [2] etc) lack either:

- Multivariate signals
- Explicit dependency graphs
- Real disruption labels

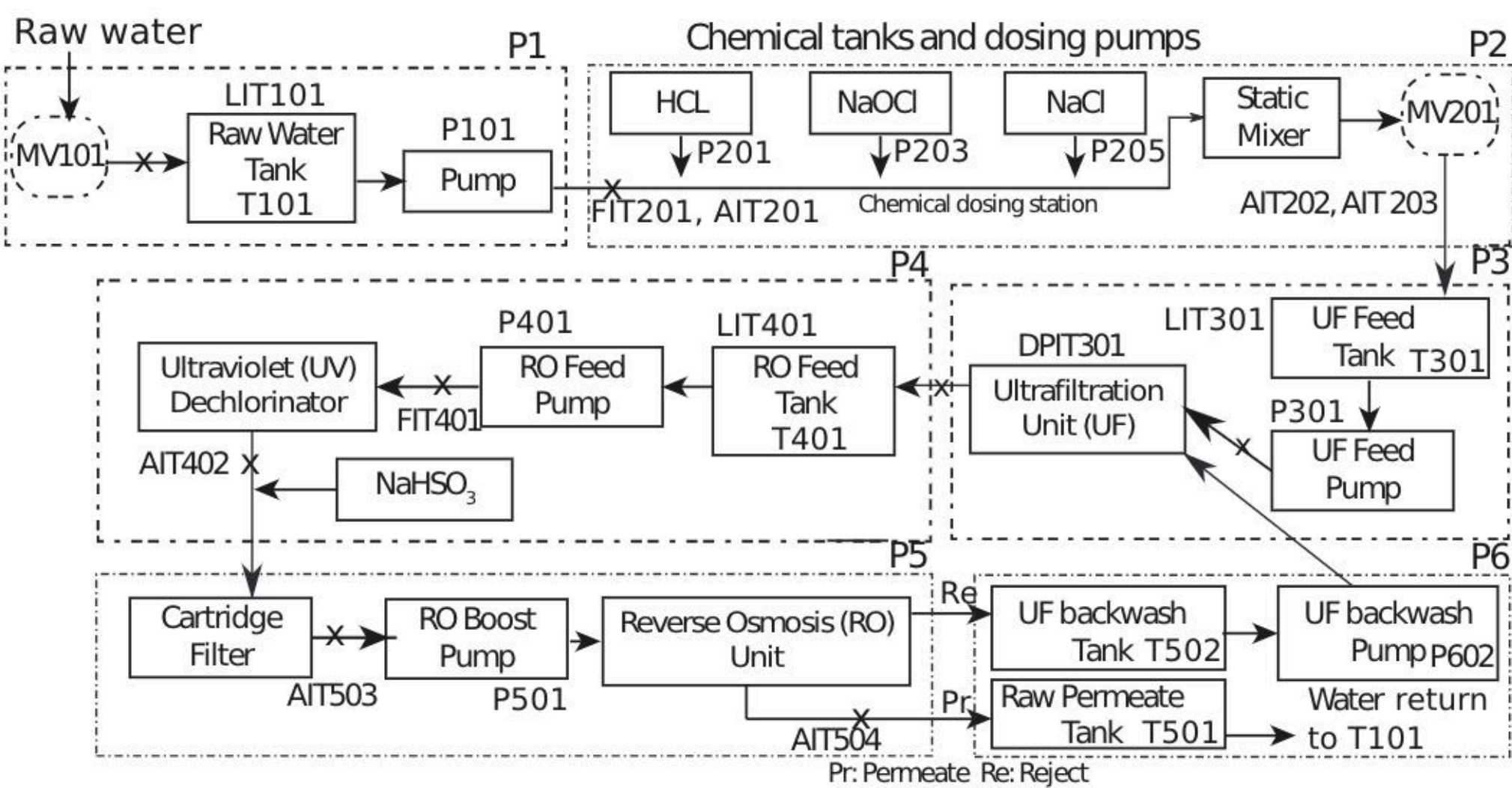
[1]:A. P. Mathur and N. O. Tippenhauer, "SWaT: a water treatment testbed for research and training on ICS security," 2016 International Workshop on Cyber-physical Systems for Smart Water Networks.

[2]: WADI: a water distribution testbed for research in the design of secure cyber physical systems. In Panagiotis Tsakalides and Baltasar Beferull-Lozano, editors, Proceedings of the 3rd International Workshop on Cyber-Physical Systems for Smart Water Networks

MOTIVATION

Industrial Benchmarks (SWaT [1], WADI [2]):

- Very simple, sequential, dependency graph.
- Synthetic anomalies, unrealistic.



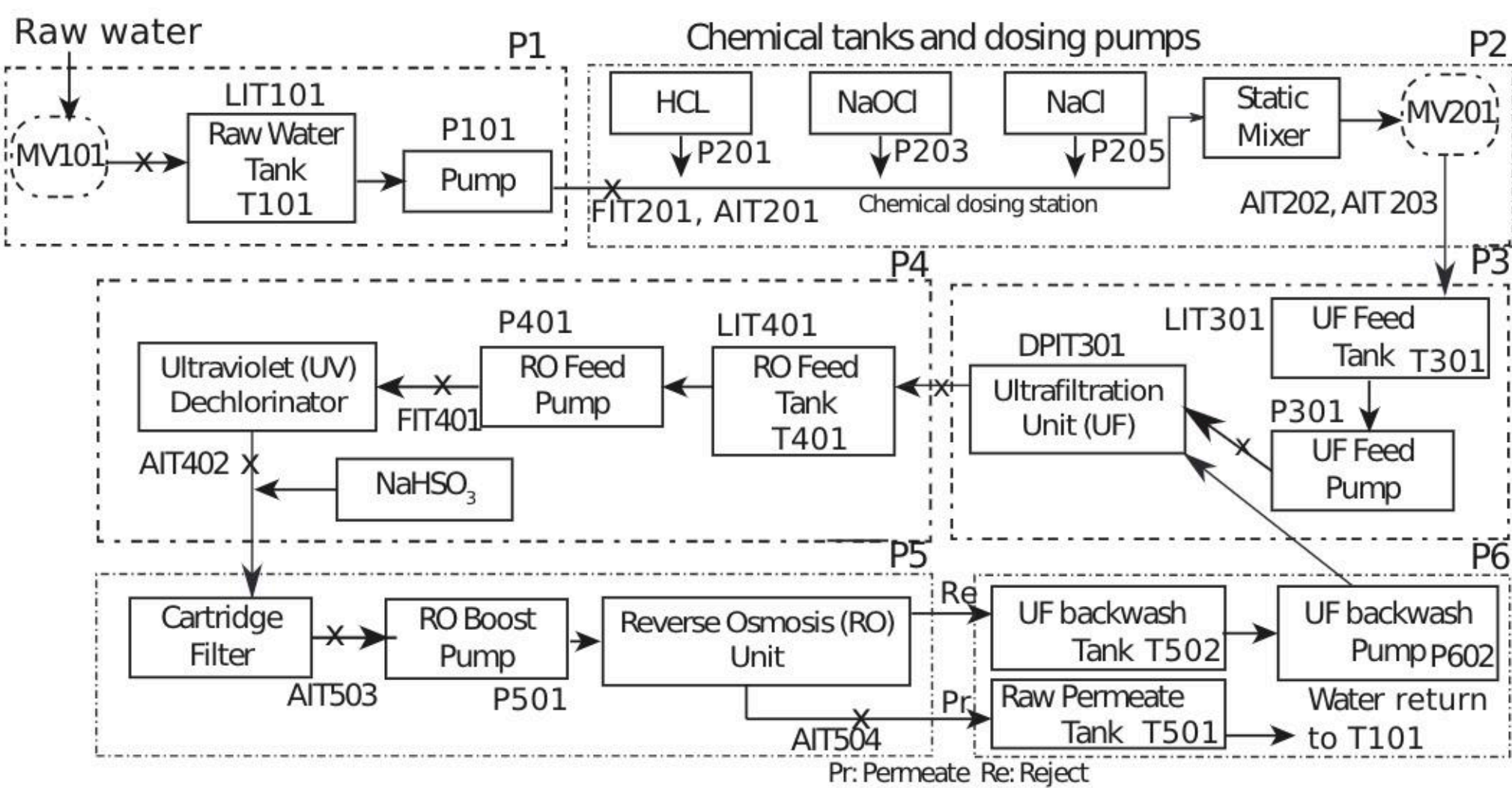
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MOTIVATION

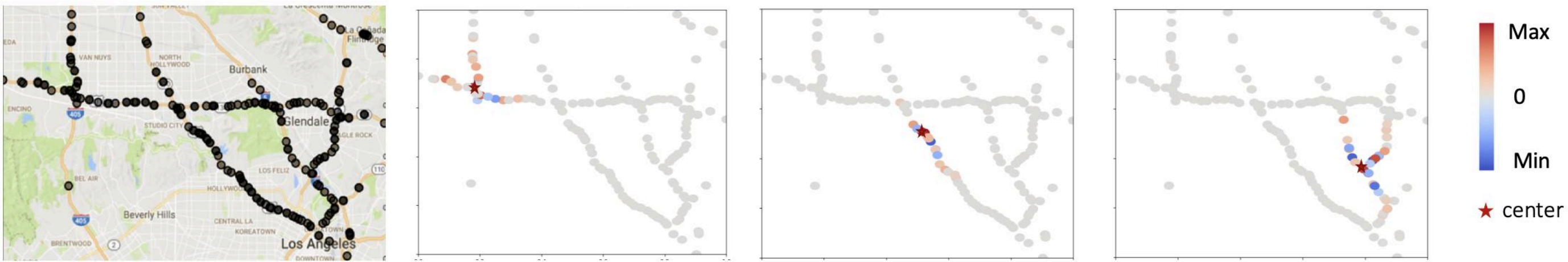
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Traffic/Air Quality Benchmarks:

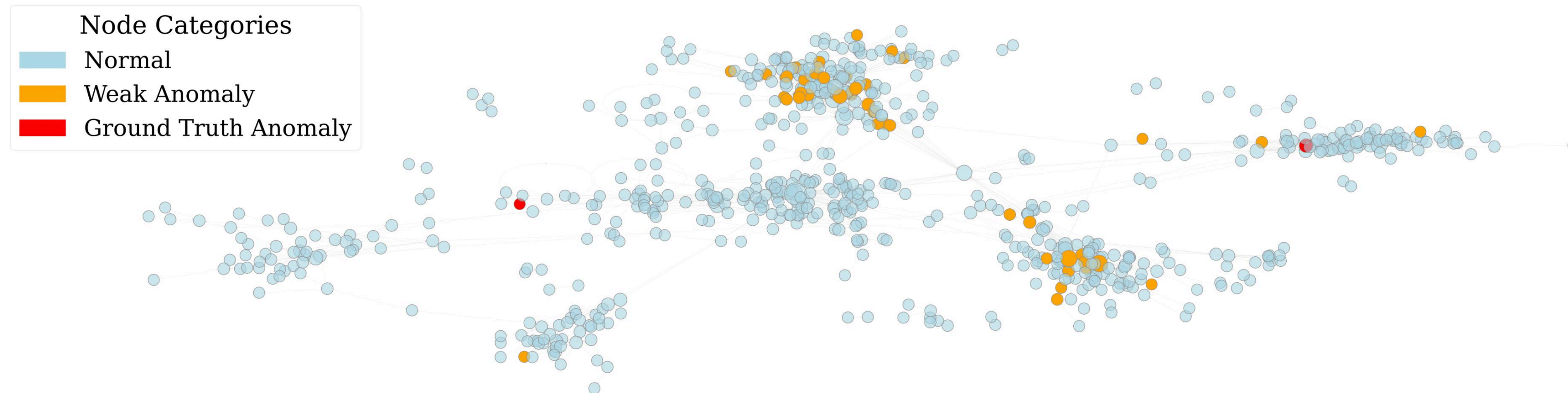
- Univariate sensor signals.
- No incident information.



[1]: A. P. Mathur and N. O. Tippenhauer, "SWaT: a water treatment testbed for research and training on ICS security," 2016 International Workshop on Cyber-physical Systems for Smart Water Networks.
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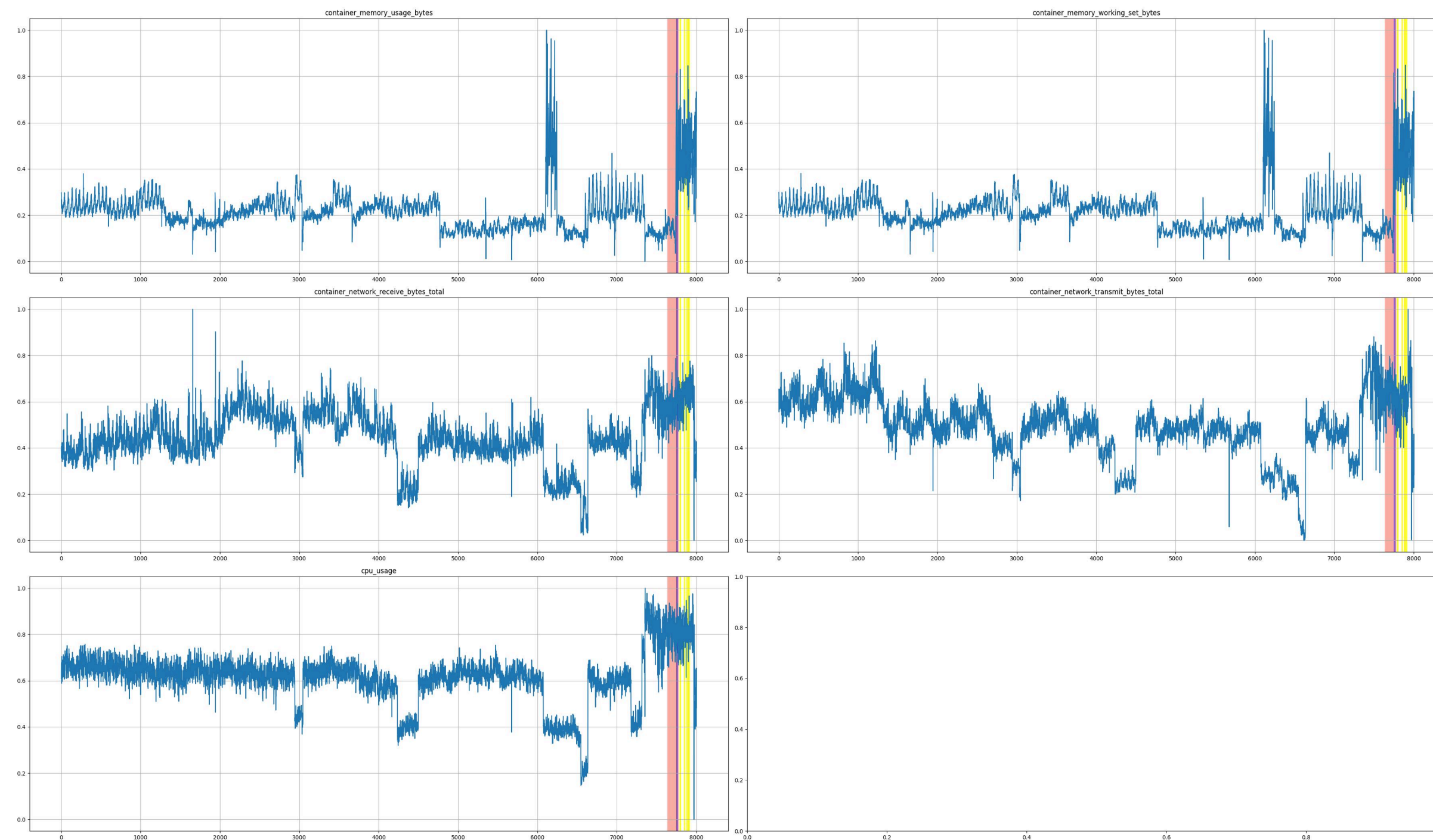
We introduce **ChronoGraph**:

- A **real-world** multivariate time series dataset derived from **six months of production** microservice telemetry, making it a difficult and relevant dataset.
- System disruption labels are flagged by Bitdefender experts, making them a very valuable resource.
 - Ground truth anomalies are confirmed disruptions.
 - Weak anomalies are disruptions detected by our algorithms.



Dataset Structure:

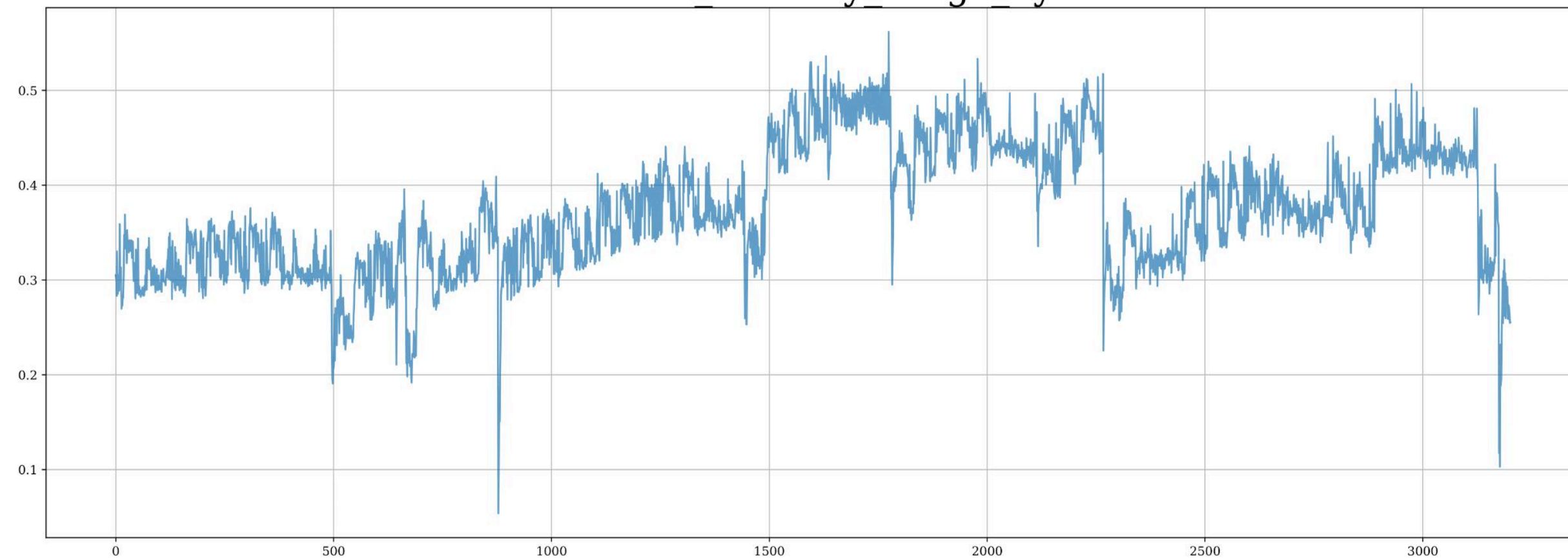
- Each **service** is a node (~**700**), represented by a 5D time series with ~**8,000** time steps per service (30 min/step).
 - First 60% steps for training, 40% for testing
- **Inter-service communication** is captured with 8D time series on **directed edges** (~**1500**).



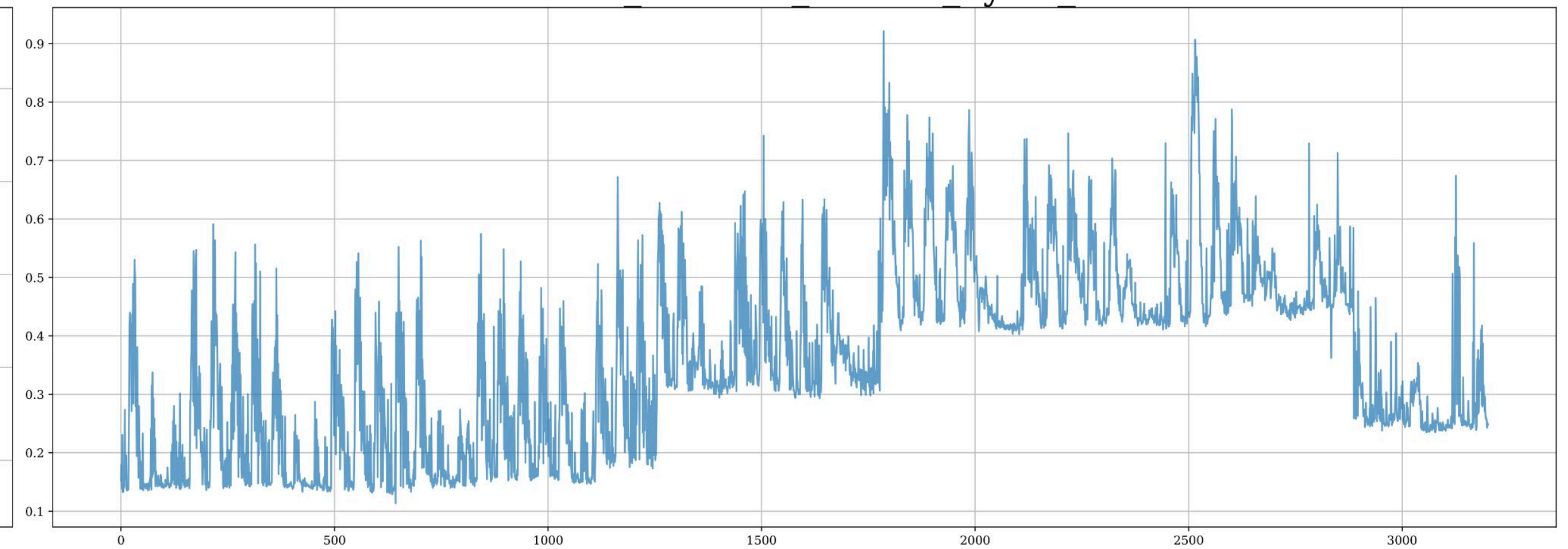
CHRONOGRAPH

- Nodes track critical performance metrics (e.g. CPU & memory usage).
- Edges track request volumes, return codes, and latency.

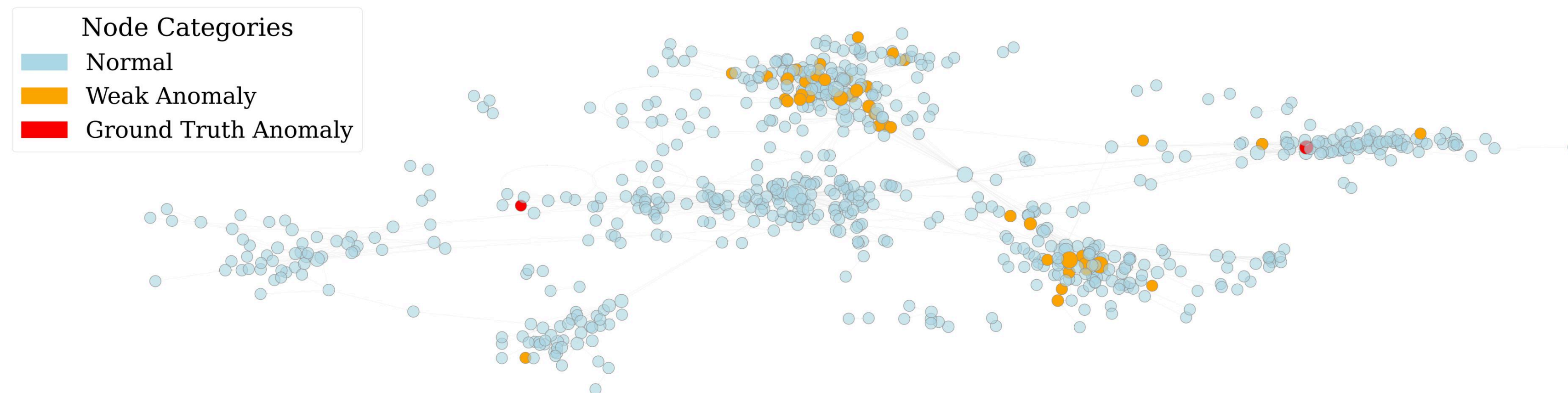
container_memory_usage_bytes



container_network_receive_bytes_total



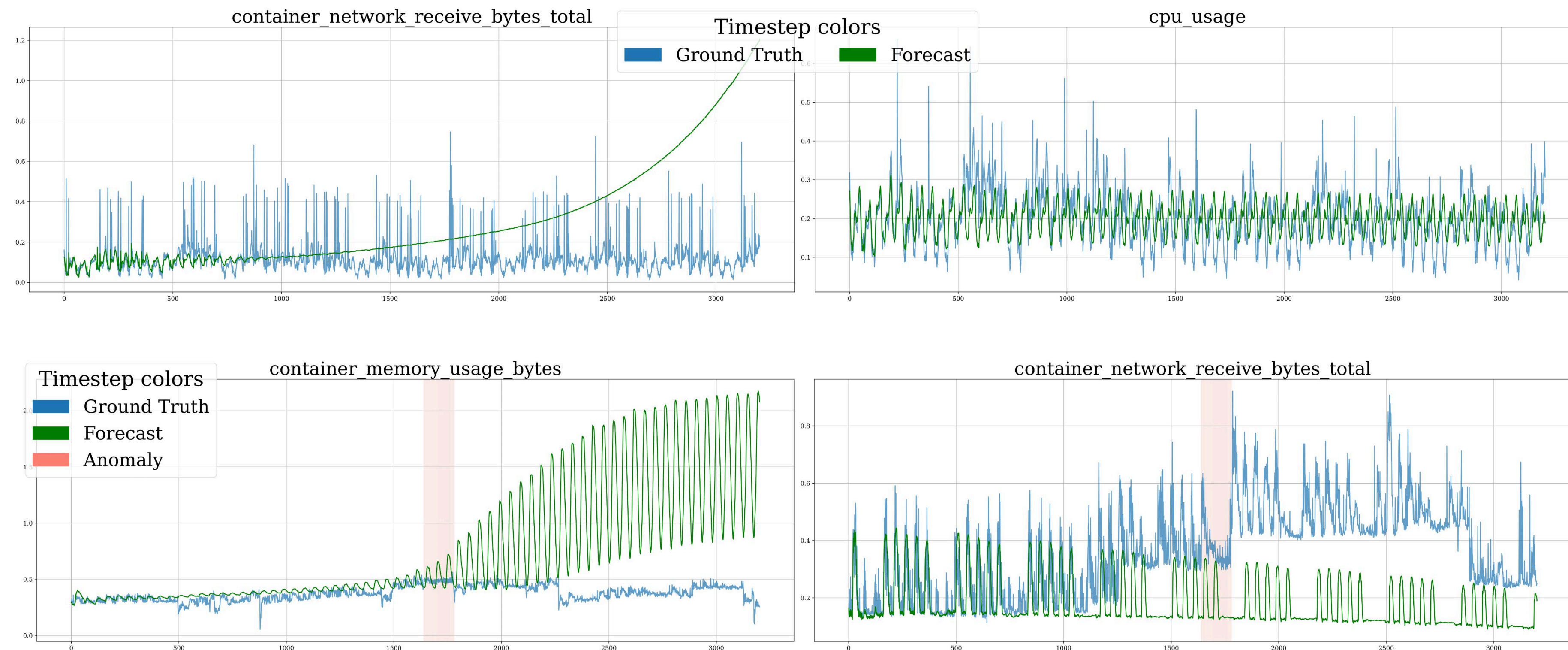
- **Graph Topology:** The service dependency graph reveals the structural layout of the microservices.
- **Incident Propagation:** Anomalies propagate along dependency paths, uncovering failure dynamics.



CHRONOGRAPH

Why ChronoGraph?

- **Model Development:** Allows developing models that capture topological correlations.
- **Evaluation Benchmark:** Evaluation of forecasting and anomaly detection approaches.



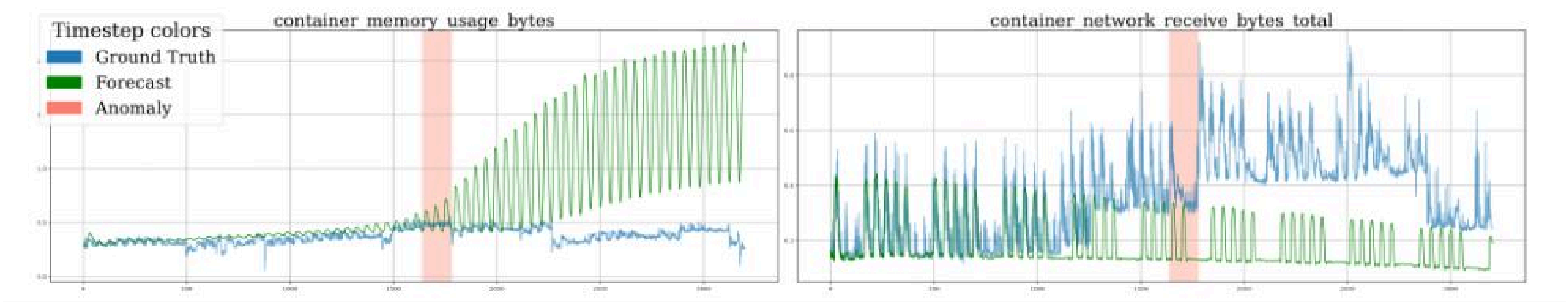
- We benchmark:
 - Forecasting methods (Prophet),
 - Time-series foundation models (TabPFN-TS, Chronos),
 - Anomaly detectors (Isolation Forest, OCSVM, Autoencoder).
- We highlight **key limitations** of current approaches:
 - Bad long-term forecasting.
 - Inability to leverage the graph structure.

Forecasting results

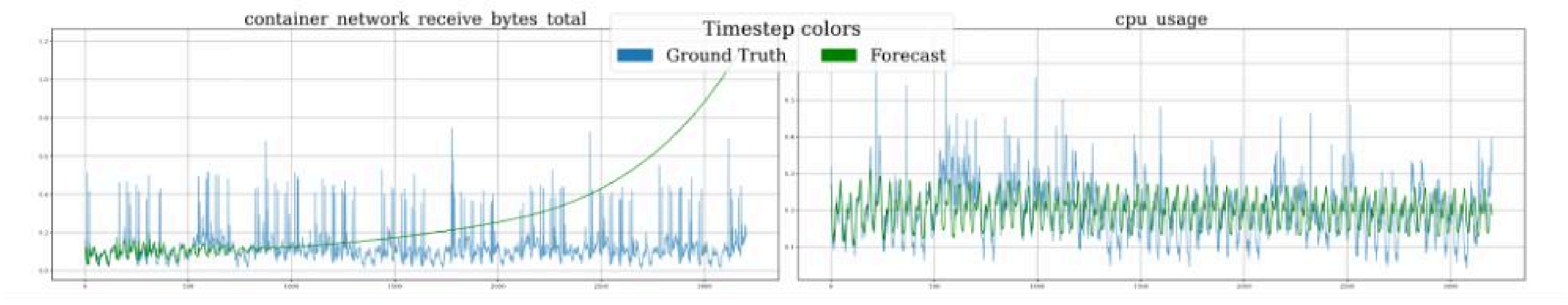
- All models achieve substantially lower errors on a shorter horizon (500 steps) than long horizons (~3000 steps).
 - TabPFN-TS most stable;
 - Chronos best short-term, worst long-term.
- Models showcase limited long-term stability

Model	window=3202			window=500		
	MAE (↓)	MSE (↓)	MASE (↓)	MAE (↓)	MSE (↓)	MASE (↓)
Prophet	0.125 ±0.067	0.044 ±0.054	7.182±11.210	0.069±0.044	0.013±0.022	3.143±3.663
Chronos	0.150±0.173	0.343±2.426	7.902±12.707	0.044 ±0.030	0.007 ±0.015	1.938 ±1.731
TabPFN-TS	0.125 ±0.125	0.089±1.172	6.205 ±9.315	0.109±0.061	0.026±0.031	5.082±11.013

CHRONOGRAPH FORECASTING



Left: prediction degrades when anomaly occurs.
Right: the model captures only a periodic baseline.



Left: network traffic is accurate at first but then drifts upward even without an anomaly, illustrating the limited stability of long-horizon predictions.
Right: CPU usage is forecast accurately across the full test time window

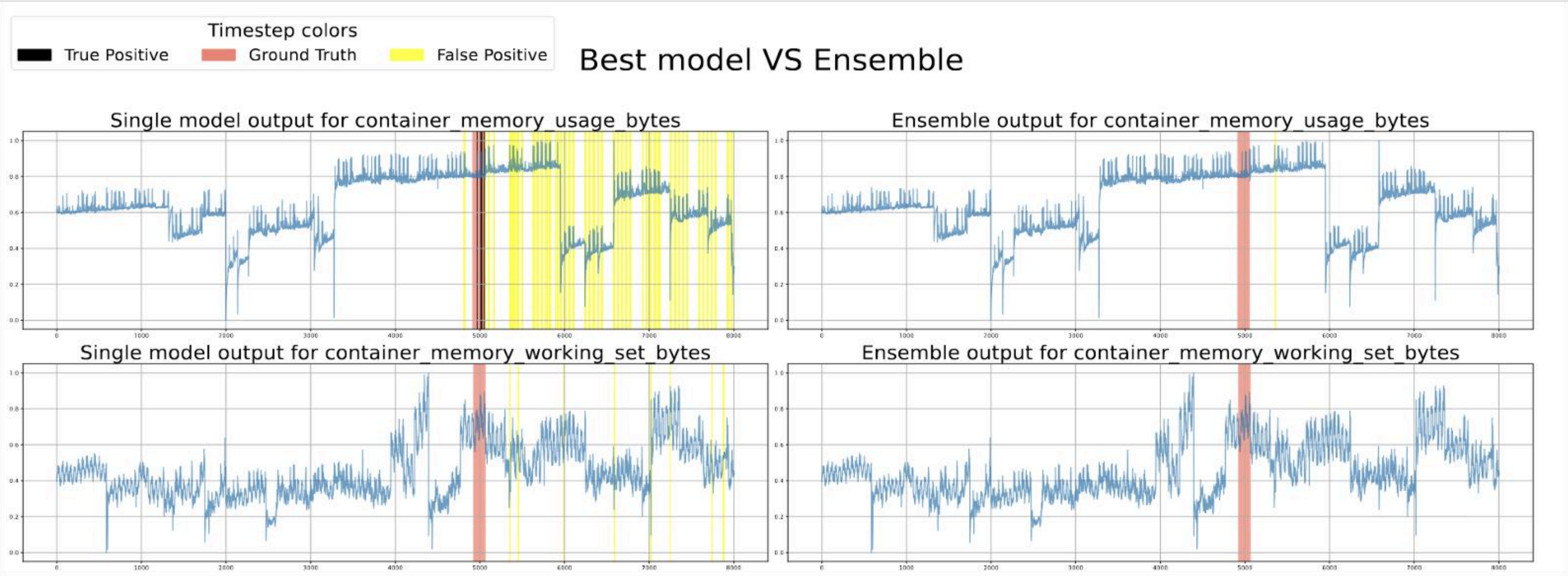
Anomaly detection (service disruptions):

- Forecasting methods (residuals for anomaly scoring) :
 - Prophet detects anomalies, but has high false negatives.
 - Foundation models weak overall.
- Isolation Forest, OC-SVM: more anomalies, many false positives.
- Ensemble balances methods, but remains limited.

Table 2: Performance of individual models and an ensemble, evaluated on human-labeled service disruptions (partial coverage). The ensemble (*) combines Prophet, Isolation Forest, and Autoencoder.

Method	$F1_K$ ($\uparrow$)	ROC_K ($\uparrow$)	FP rate ($\downarrow$)	FN rate ($\downarrow$)	$F1$ ($\uparrow$)
Prophet	20.57	62.97	2.02	97.98	2.39
Isolation Forest	17.49	56.39	46.9	50.48	7.08
OC-SVM	14.46	54.31	22.13	77.08	5.50
Autoencoder	13.86	59.79	0.38	99.58	0.72
TabPFN-TS	12.37	54.08	0.55	99.79	0.31
Chronos	12.41	49.78	2.49	97.84	2.49
Ensemble*	16.92	60.95	0.20	99.58	0.73

Anomaly detection (service disruptions)



How do we proceed?

- Exploit the graph:** graph topology might play a crucial role for both anomalies and forecasting.
- Initial results combining message-passing with MTS forecast models seem promising.

Could **Temporal Graph Learning** be a good inspiration?

- Both multivariate and temporal graph learning communities work with temporal data.
- Expertise from both areas might provide further insights.

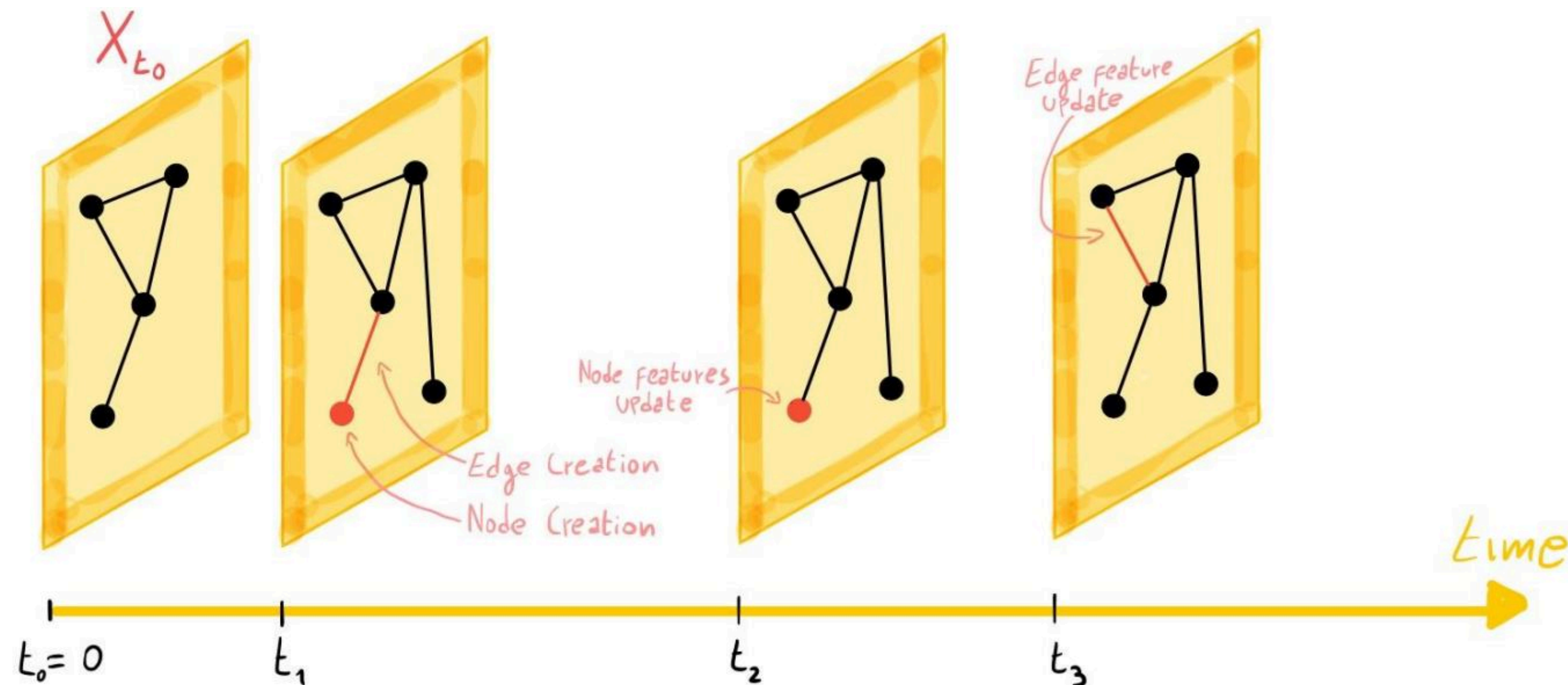


Figure source: <https://towardsdatascience.com/temporal-graph-learning-in-2023-d28d1640dbf2/>

We hope that **ChronoGraph** will:

- **Improve** evaluation of anomaly detection methods.
- **Catalyze** research towards **graph-aware forecasting** in multivariate time series.
- **Connect** the multivariate time series and temporal graph learning communities by providing a shared benchmark.

Key Insights:

- Forecasting models handle short horizons but fail long-term.
- Anomalies can significantly affect forecasts.
- Detected anomalies cluster along connected services → propagation effects.

Future Work:

- Graph-aware forecasting models.
- Graph-based anomaly detection to capture disruption spread.
- Further predict and analyze weak anomalies together with experts.

Thanks!

Thanks!



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